

Date	03 October 2026
Team ID	6ab8f5efc23107aaaa7325ec
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Project Design Phase-II

Technology Stack (Architecture & Stack)

Technical Architecture:

The solution automates the classification, prioritization, and routing of support tickets using a low-code Flow Designer platform (ServiceNow Flow Designer / Microsoft Power Automate) combined with AI classification capabilities. The architecture follows a modern event-driven, integration-centric design.

High-Level Architecture Flow:

- Ticket Created → Record Created trigger fires in Flow Designer
- Flow extracts Short Description + Description (+ optional attachments)
- Hybrid Classification Engine:
 - Rule-based / Keyword matching (high confidence cases)
 - AI Classification Model (Predictive Intelligence / AI Builder / External LLM)
- Outputs: Category, Subcategory, Priority, Assignment Group, Confidence Score, Explanation
- Decision: If confidence ≥ threshold → Auto-update ticket & assign
 Else → Route to Manual Review queue + notify agent
- Agent feedback captured for continuous model improvement
- All decisions logged for audit & analytics

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Agents interact via Service Portal / Agent Workspace to view classified tickets, override classifications, and provide feedback.	ServiceNow Service Portal / Agent Workspace (or Power Apps)
2	Application Logic-1 (Orchestration)	End-to-end flow that triggers on ticket creation, calls classification, applies updates, and handles routing decisions.	ServiceNow Flow Designer (or Microsoft Power Automate)
3	Application Logic-2 (Rule Engine)	Keyword / Regex / Decision Table based classification for high-confidence and common ticket patterns.	Flow Designer Decision Tables / Script Steps / Business Rules
4	Application Logic-3 (AI Classification)	Machine learning / Generative AI model that classifies free-text ticket content into Category, Priority and Assignment Group.	ServiceNow Predictive Intelligence / AI Builder Text Classification / Azure OpenAI / Custom ML model
5	Database	Stores tickets, classification history, feedback, confidence scores and audit logs.	ServiceNow Platform Database (MariaDB) or Dataverse / SQL
6	Cloud Database / Logging	Optional external store for model training data and analytics dashboards.	IBM Cloudant / Azure Cosmos DB / ServiceNow Performance Analytics
7	File Storage	Temporary storage for any ticket attachments used in classification (if Document Intelligence is used).	ServiceNow Attachment Storage / Azure Blob / Local filesystem
8	External API-1	Optional call to external LLM or custom classification endpoint when platform native AI is insufficient.	REST / Integration Hub / HTTP Connector
9	External API-2	Optional CMDB / User directory lookup to enrich priority (e.g., VIP user, critical CI).	ServiceNow CMDB API / Microsoft Graph / LDAP
10	Machine Learning Model	Text classification model trained on	ServiceNow Classification

		historical tickets to predict Category + Assignment Group + Priority.	Framework / AI Builder Custom Model / Azure ML / scikit-learn / Hugging Face
11	Infrastructure	Platform runs on SaaS cloud. Flow Designer executes server-side. No dedicated servers required for core solution.	ServiceNow Cloud (or Microsoft Power Platform / Azure)

Table-2: Application Characteristics

S.No	Characteristics	Description / Technology
1	Open-Source Frameworks	Flow Designer is low-code (proprietary platform). Optional custom ML models can use open-source libraries (scikit-learn, Transformers, spaCy). Integration via REST.
2	Security Implementations	Platform RBAC / ACL, encrypted data at rest & in transit (TLS), audit logging of every classification decision, least-privilege service accounts for integrations, OAuth/API keys for external AI.
3	Scalable Architecture	Event-driven Flow Designer scales with platform. Stateless classification steps. Can handle thousands of tickets/hour. Horizontal scaling managed by the SaaS platform.
4	Availability	Platform SLA typically 99.9%+. Flow retries on transient failures. Fallback to Manual Review queue ensures tickets are never lost.
5	Performance	Classification target < 3–5 seconds per ticket. Caching of common patterns. Asynchronous processing for non-critical paths. Performance Analytics dashboards for monitoring.

References:

<https://c4model.com/>

<https://docs.servicenow.com/bundle/xanadu-build-workflows/page/administer/flow-designer/concept/flow-designer.html>

<https://learn.microsoft.com/en-us/ai-builder/text-classification-model-in-flow>

<https://www.ibm.com/cloud/architecture>